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Object frequency is underestimated in large-scale scene image databases

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Why is object frequency important?

- **Experiment design** and **stimulus choice**
  - Identify objects frequently occurring in a scene category
  - Match object frequency across categories or conditions
  - Ensure representativeness of objects

- **Statistical control**
  - Account for object frequency after experiment is run

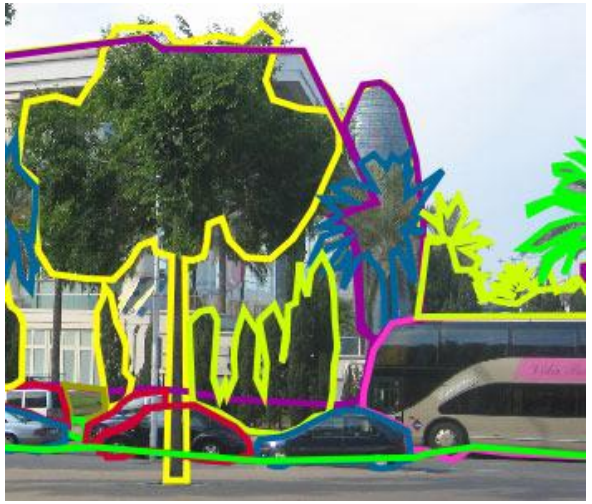
- **Theoretical interest**
  - Object frequency is a diagnostic object property for scene categorization (Wiesmann & Võ, *Sci. Rep.*, 2023)

Previous measurement approaches

- **Researcher intuition & human ratings**
  - Arguably biased, suggested to overestimate object frequency (Greene, *Cognition*, 2016)

- **Labelled scene image databases: LabelMe, ADE20K…** (Russell et al., *Int. J. Comput. Vis*, 2008; Zhou et al., *CVPR*, 2017)
  - Supposedly more objective than human intuition/ratings (Greene, *Cognition*, 2016)

- **Our approach: Real-world object occurrence statistics**
  - Have human participants *count* objects in their environment



Measuring real-world object frequency

Online questionnaire

- ***N* = 151** (mostly German undergraduate students)
- **79 objects** from different sources (Wiesmann & Võ, *Sci. Rep.*, 2023; ADE20K, Zhou et al., *CVPR*, 2017; Greene, *Front. Psychol.*, 2013)
- **4 scene categories:** Kitchen, bathroom, bedroom, living room (+ 'other')

- For each object…
  - **Count number of object instances** in each room (estimate when > 10)
  - Rate **object visibility** in each room (5-point Likert scale)
  - Rate **frequency of object usage** in each room (6-point Likert scale)

Measuring real-world object frequency

Frequency: Probability of finding object X in a scene of category Y $P(Object|Scene\ category)$

(Greene, *Front. Psychol.*, 2013; Wiesmann & Võ, *Sci. Rep.*, 2023)

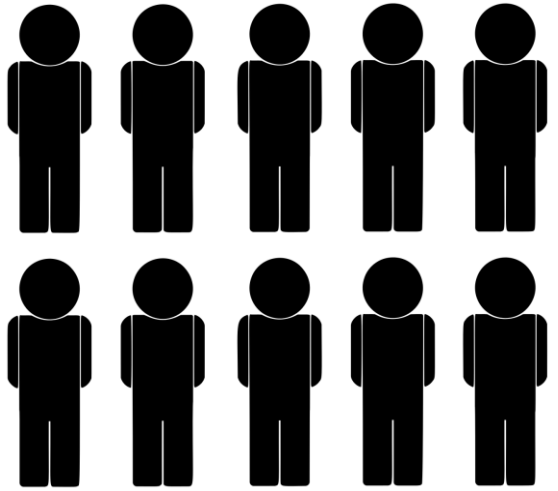
Scene image databases



...

40/100 bathroom images contain (at least) one soap → $P(Soap|Bathroom)$ = .40

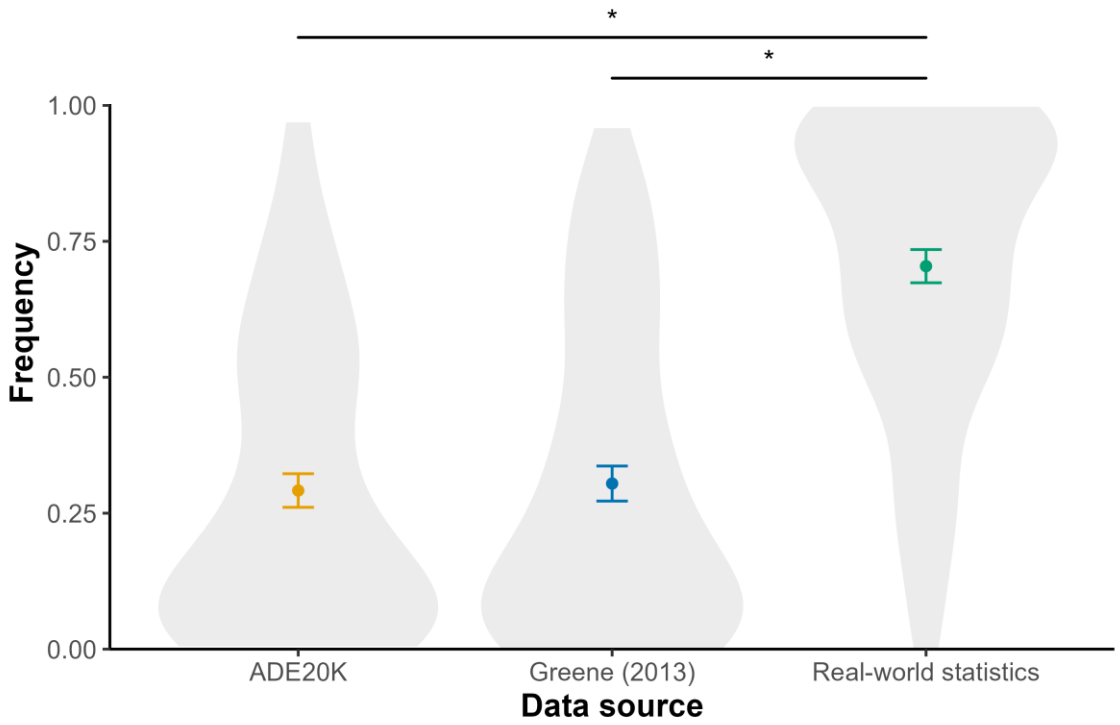
Real-world statistics (RWS)

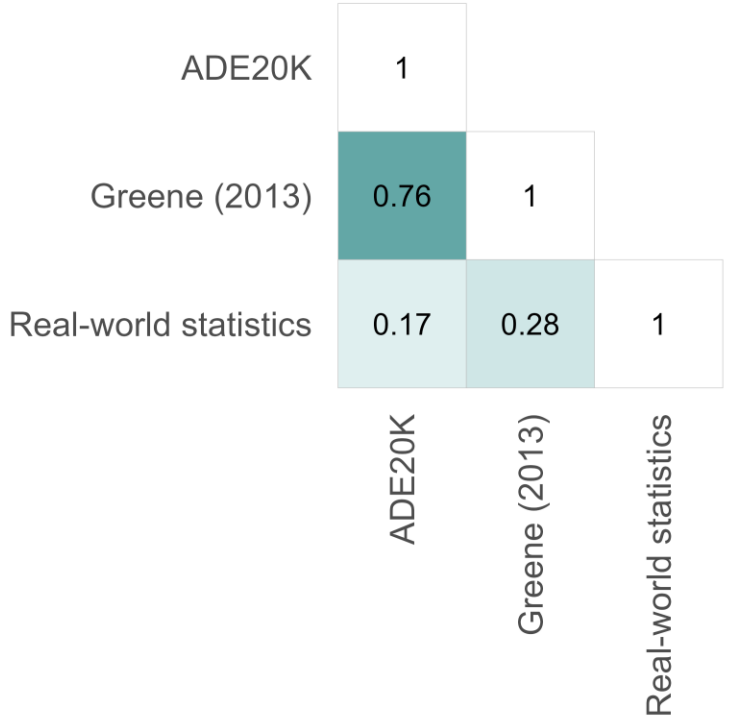


62/100 participants have (at least) one soap in their bathroom → $P(Soap|Bathroom)$ = .62

How do databases compare to the real world?

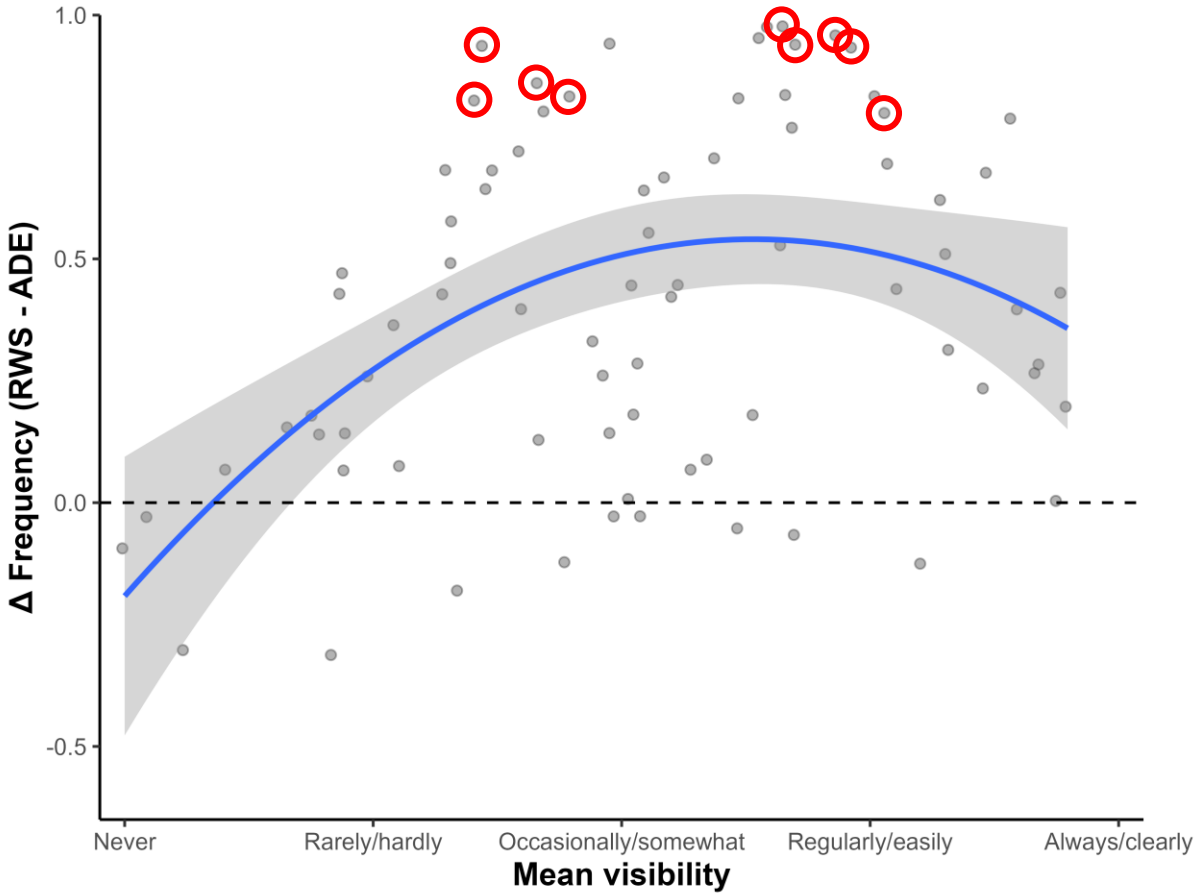
Underestimation of object frequency





→ Objects are **more than twice as likely to occur in the real world** compared to database predictions!

Which objects are underrepresented?





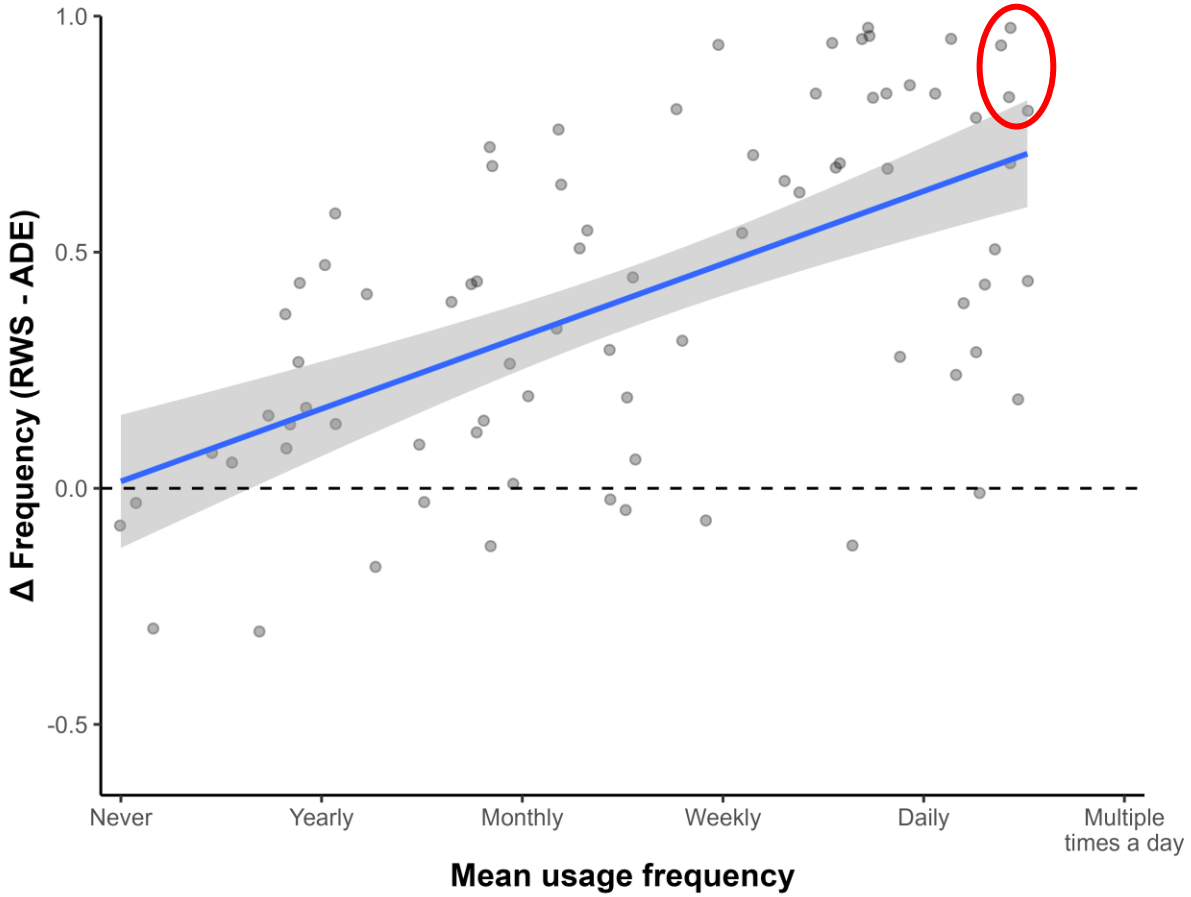
ADE: .005 .030 .003 .035 .210



ADE: .004 .098 .142 .160

→ Underrepresentation of objects that are **not always visible** in everyday life

Do these objects matter?





→ Underrepresented objects are **used frequently** in everyday life!

Why are objects underrepresented in databases?

Image selection bias (Torralba & Efros, *CVPR*, 2011; Greene et al., *arXiv*, 2025)





Name That Dataset! (Torralba & Efros, *CVPR*, 2011)

Why are objects underrepresented in databases?

Hidden object bias (see also David & Võ, *VSS*, 2022)





Why are objects underrepresented in databases?

Photographer & aesthetics bias (Torralba & Efros, *CVPR*, 2011; Greene et al., *arXiv*, 2025)



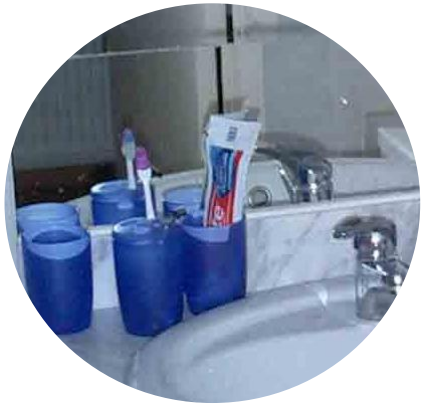




Why are objects underrepresented in databases?

Labelling bias (Torralba & Efros, *CVPR*, 2011; Greene et al., *arXiv*, 2025; Haliburton et al., *AI Ethics*, 2025)









Why are objects underrepresented in databases?

Culture & socio-economic bias (Torralba & Efros, *CVPR*, 2011; Greene et al., *arXiv*, 2025; Haliburton et al., *AI Ethics*, 2025)

Most overrepresented objects:







Limitations

- Primarily **WEIRD student sample** (Henrich et al., *Behav. Brain Sci.*, 2010)
- **Small sample size** (*N* = 151)
  - But compare smaller number (< 120) of scenes per category used in Greene (*Front. Psychol.*, 2013)
- Currently **limited to private indoor scenes**

Conclusions

- Scene image **databases underestimate object frequency** in the real world!
- Especially **smaller objects** that are usually **hidden/stored away** are underrepresented
- **Databases are biased:** Image selection, hidden object, labelling, photographer, aesthetics, cultural & socio-economic biases
  - Important not only for object frequency estimation but also **training of models**

Suggestions

- Choose object frequency measure carefully for the **purpose of your project**
- Database measures may be appropriate when making **statements about database images**
- Consider **assessing real-world object frequency explicitly** to get a more realistic measure

Thanks!

Go see Lea's poster tomorrow at 10!







Feron Y. Basoeki

Melissa L.-H. Võ

Esin Gün Gürsoy

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